



APPLIED AI | NLP | DOCUMENT AUTOMATION

Bruno Ferreira

Applied AI Engineer / AI Solutions Engineer

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PROFILE Mid-level Applied AI Engineer with 4+ years across applied research, client delivery and university teaching. I build NLP, language-model and document-automation solutions, from process discovery and implementation to evaluation, human validation and team training. I also provide AI decision consulting, including deciding what is worth automating and what is not. Based in Portugal, I am open to contract projects and mid-level roles across the European Union.

Applied AI | NLP and language models | Document automation | AI evaluation | Full-stack delivery

01 / PROFESSIONAL EXPERIENCE

Independent Applied AI Consultant and Trainer

June 2025 - Present

Selected client work | Portugal and remote

- Take direct responsibility for process discovery, requirements, solution architecture, implementation, testing and technical documentation within each agreed project scope.
- Advise on use-case scope by mapping workflows, identifying what is worth automating, what should remain manual, and the evidence and acceptance criteria needed to proceed.
- Design and deliver practical AI training for client and developer teams, tailored to the workflows and tools in use.

Invited Lecturer

February 2024 - Present

University of Coimbra

- Modules taught: Principles of Databases, Fundamentals of Artificial Intelligence, Principles of Java, Principles of C, Natural Language Processing and Knowledge Representation.
- Preparation of learning materials, guidance and assessment of student work, and supervision of MSc dissertations.

Researcher

February 2022 - June 2025

CISUC | Centre for Responsible AI

- Conducted research in NLP, NLG, Transformers, Information Extraction, Question Answering, language models, Explainable AI, causal inference and model compression.
- Published and presented work on Question Answering, information extraction and the evaluation of knowledge encoded in language models.

02 / SELECTED APPLIED AI PROJECTS

AI platform for US visa workflows

2025 - Present

CONTEXT: Multi-tenant LegalTech platform that applies USCIS criteria to scattered documentation in a consistent, evidence-guided workflow. **MY CONTRIBUTION:** Designed and implemented workflow components for structured intake, triage, dynamic questionnaires, document management, OCR, evidence-guided language-model analysis and legal-review reports within a multi-tenant full-stack architecture.

TOOLS: Python, TypeScript, FastAPI, Next.js, PostgreSQL, Redis, Alembic, Docker, OCR, LLMs, ClamAV, Sentry

HVAC technical-specification extraction

2026

CONTEXT: Pipeline that transforms Portuguese HVAC specification PDFs into structured Excel and JSON outputs for engineering review. **MY CONTRIBUTION:** Designed and implemented a schema-driven hybrid extraction workflow combining embedding-based RAG, deterministic table mapping and regular expressions with vision and language models, preserving source-page evidence and benchmarking against manual ground truth.

TOOLS: Python, RAG, embeddings, regular expressions, PDF and table extraction, vision, LLMs, Excel, JSON, XAI

Quality Copilot for document validation

2026

CONTEXT: Specialised modules that support comparison and validation of packaging and technical documentation. **MY CONTRIBUTION:** Designed and built workflows for dimensional checks, content validation, correction verification and technical-sheet preparation, defining evidence requirements and human-review gates for every stage.

TOOLS: Python, OCR, Claude Code skills, document automation

Global THE 17.3 evidence evaluation

2025

CONTEXT: Commissioned to design and carry out a systematic review of public university information for THE Impact Rankings indicators 17.3.1 to 17.3.17. **MY CONTRIBUTION:** Defined the evidence-review method and a repeatable assessment process, applied consistently across the 17 indicators.

TOOLS: Evidence review, evaluation design, structured analysis

Privacy-by-design AI platform

Product in development

CONTEXT: Own product combining local personal-data detection and anonymisation with external language-model workflows. **MY CONTRIBUTION:** Defined the product strategy and architecture and built a functional PWA with document processing, model-assisted chat, projects, user profiles and verifiable privacy controls.

TOOLS: React, TypeScript, NER, ONNX, LLMs, Privacy by Design

03 / TECHNICAL CAPABILITIES

Applied AI: NLP, language models, Information Extraction, Question Answering, OCR, XAI, prompting, evaluation and human review

Engineering: Python, TypeScript, React, Next.js, FastAPI, SQL, PostgreSQL, APIs, Git, Unix, Docker and document automation

From problem to delivery: Needs discovery, prototyping, testing, technical review, documentation, deployment preparation and training

Development with AI tools: Claude Code, Codex, OpenCode and Pi for implementation, review, testing and documentation, with defined review criteria and my own supervision

04 / EDUCATION

PhD in Informatics Engineering

2022 - Present

University of Coimbra

MSc in Informatics Engineering - Intelligent Systems

2022

University of Coimbra

BSc in Informatics Engineering

2021

University of Coimbra

05 / SELECTED PUBLICATIONS

- Towards Automated Evaluation of Knowledge Encoded in Large Language Models | LREC-COLING, 2024
- BATS-PT: Assessing Portuguese Masked Language Models | PROPOR, 2024
- Leveraging Question Answering for Domain-Agnostic Information Extraction | CIARP, 2023
- Evaluating the Extraction of Toxicological Properties with Extractive Question Answering | EANN, 2023
- SafetyDesk: Accelerating the Production of Toxicological Reports | EXP.AT, 2023
- ETFs Information Extraction from KIIDs | Environmental Finance for the Common Good, 2023
- Question Answering for Toxicological Information Extraction | SLATE, 2022
- Safety Desk: Extraction and Analysis of Textual Information to Build a Reporting System | MSc dissertation, 2022

06 / LANGUAGES

Portuguese - Native | English - Advanced | Spanish - Good comprehension, basic speaking

07 / WAYS TO WORK TOGETHER

Contract projects | In-company training | AI decision consulting | Startup support | Mid-level EU roles

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